

$\mathbb{R}^{2k}$ is Theoretically Large Enough for Embedding-based Top- k Retrieval

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Geometric approximability is not the obstruction.

MED = Minimal Dimension to achieve (robust) k -shattering

k -shattering

Store m objects as vectors. For every answer set $S \subseteq X$, $1 \leq |S| \leq k$, choose a query and threshold:

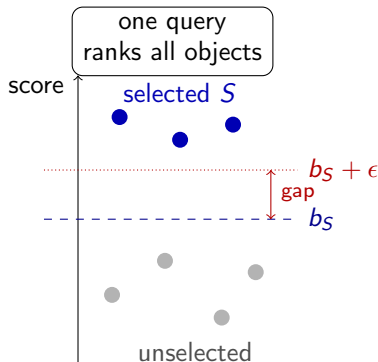
$$\min_{i \in S} s(v_i, u_S) > b_S \geq \max_{j \notin S} s(v_j, u_S).$$

If $|S|$ is known, this is top- $|S|$ retrieval.

Robust k -shattering

Unit-normalized objects and queries must also have margin:

$$\min_{i \in S} s(v_i, u_S) \geq \max_{j \notin S} s(v_j, u_S) + \epsilon.$$



Exact k -shattering asks for strict separation. Robust k -shattering asks for a normalized score gap that cannot be removed by rescaling.

MED bounds

Upper bound construction: cyclic polytope and square polynomials

Pick $t_1 < t_2 < \dots < t_m$. Put objects in $\mathbb{R}^{2k}$:

$$v_i = (t_i, t_i^2, \dots, t_i^{2k}).$$

For target S , define

$$P_S(t) = \prod_{i \in S} (t - t_i).$$

Use coefficients of $-P_S(t)^2$ as u_S .

Then selected objects score 0, and every unselected object scores < 0 .

Lower bound: VC dimension

MED realizes every subset of any k chosen objects, so the score threshold class shatters k points.

Linear thresholds in $\mathbb{R}^d$ have VC dimension $d + 1$, hence

$$d + 1 \geq k \quad \Rightarrow \quad d \geq k - 1.$$

Scoring	Lower	Upper
Inner product	$k - 1$	$2k$
Euclidean	$k - 1$	$2k$
Cosine	$k - 1$	$2k + 1$

Exact MED is $\Theta(k)$ and does not grow with m .

RMED: feasible margins and Gaussian upper bound

Feasibility of ϵ (informal).

Every robust witness must satisfy

$$\epsilon \leq \epsilon_{\star}(m, k) = \frac{m}{\sqrt{k(m-1)(m-k)}}.$$

- $\epsilon > \epsilon_{\star}(m, k)$: infeasible.
- $\epsilon \leq \epsilon_{\star}(m, k)$: feasible in $\mathbb{R}^{m-1}$ by a regular simplex.
- If $m/k \rightarrow \infty$, then $\epsilon_{\star}(m, k) \sim 1/\sqrt{k}$.

Gaussian centroid witness.

We sample random unit object vectors v_i and construct query vectors

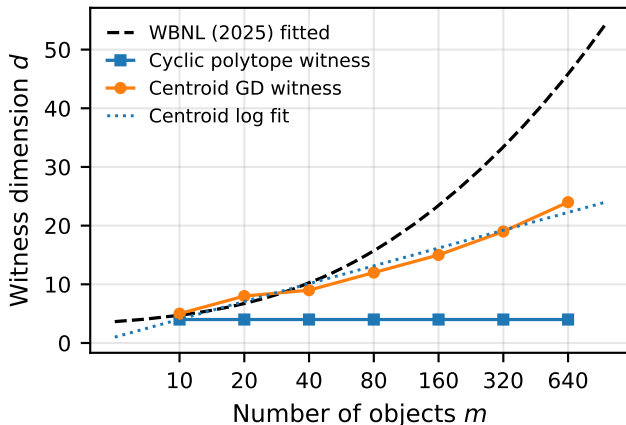
$$u_S = \frac{\sum_{i \in S} v_i}{\|\sum_{i \in S} v_i\|_2}.$$

At the feasible margin scale $\epsilon_k = c/\sqrt{k}$, when $d = Ck^2 \log m$, concentration and union bounds implies that robust MED exists.

$$\text{RMED-C}(m, k, \epsilon_k) \leq O(k^2 \log m),$$

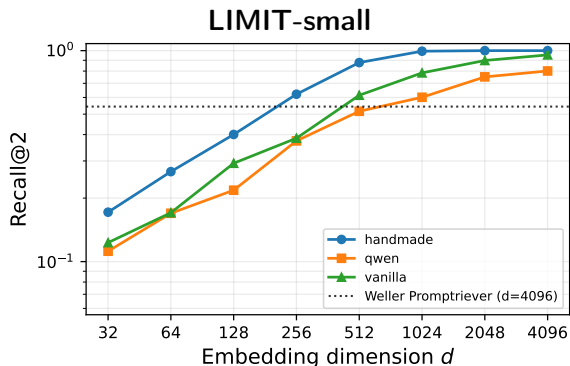
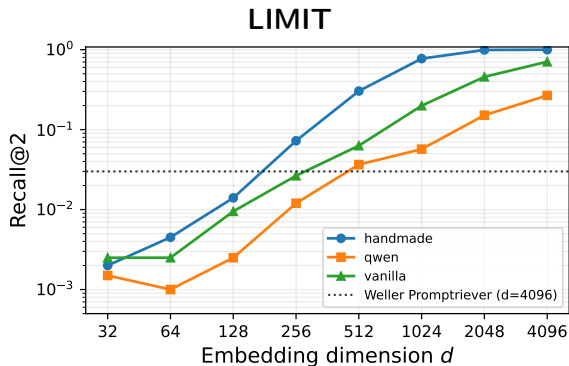
and RMED is no larger than this restricted centroid-query quantity.

Experiments (1): synthetic top-2 query



- For $k = 2$, the construction predicts an exact $d = 4$ witness.
- Centroid optimization grows slowly on the tested grid.
- These are checked upper-bound witnesses, not certified minima.

Experiments (2): LIMIT and LIMIT-small



Random additive single-vector embeddings cross the reported Promptriever line. At $d = 4096$, vanilla reaches Recall@2 0.7060 on LIMIT and 0.9545 on LIMIT-small without using labels.

Conclusion

Exact MED

Cyclic-polytopal neighborliness gives explicit $\Theta(k)$ -dimensional witnesses for arbitrary top- k answer sets.

Robust MED

Normalized margins change the regime: feasible gaps are capped by $\epsilon_{\star}(m, k)$, and Gaussian centroids give $O(k^2 \log m)$ dimensions at the $c/\sqrt{k}$ scale.

Empirical interpretation

The observed failures are not exact-capacity failures. The remaining difficulties lie in learning, tokenization, objectives, conditioning, finite precision, and optimization.